

Supplementary Information: IRSpectra-Bench — protocol, scale tables, cases, and reproducibility

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Companion to the ICLR 2027 submission

Molecular Structure Elucidation with Frontier Models:
A Benchmark on Literature-Reported IR, ^1H , and ^{13}C NMR Peak Lists

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This supplement records the blind protocol, scoring identities, scale tables, worked cases, solver-facing contracts, and reproducibility notes that exceed the ICLR main-text limit. Headline generation on the frozen $n=500$ roster is 227/500 (45.4%) top-1 and 249/500 (49.8%) recall; self-ranking precision | recall is 227/249 (91.2%). Independent forward verification of the same pools yields 204 verified / 45 misranked / 251 never proposed (`data/fverify_n500/WALL_n500.md`). Confidence intervals are not reported for $n=500$. The $n=519$ pooled print (243/519) is not the headline. Sources: `docs/HEADLINE_N500_2026-09-20.md` (spectro-agent 48fcd30) and `docs/NIGHT_POOL_2026-09-20.md`. Unprefixed figure and listing numbers refer to the main manuscript.

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1 Cross-reference to the main manuscript

Table S1: Main-text objects and their counterparts in this supplement.

main label	object	role here
Fig. fig1	Fig. 1 (fig1_leadoverview)	Protocol and $n=500$ forward verification: 249/500 recall $\times$ 204/249 verified pool = 40.8% verified; wall 204/45/251. Reprinted as Fig. S1.
Fig. fig-chemspace	Fig. 2	Literature-cohort chemistry (not reprinted).
Listing lst:payload	Listing 1 (live lstlisting + PubChem 3D)	IRexp JSON+molecule (ILJNJJNKEOAREX; CID 57398578; C ₁₂ H ₁₁ BrN ₄). SMILES and InChIKey are resource fields, not solver payload. Reprinted below.
Fig. fig-wall	appendix (fig_wall_diagnostic)	$n=500$ forward-verify wall 204/45/251. Reprinted as Fig. S2.
Fig. fig-robustness	Fig. 4	Formula-only, recency, and four-vendor controls (not reprinted).
Fig. fig-mechanism	Fig. 5	Locked v3-R25 verify-save. Reused in §9.
Fig. fig3-method	Fig. 6	Inference ladder on the 60-compound arm (not reprinted).
App. cases	Table tab:cases	Same integers as §9.

Main-text Eqs. (1)–(7) (factorisation, InChIKey-14, top-1, recall@ k , prec.|recall, ¹³C chamfer, pooled estimator) are restated in §4 with the same meaning.

2 Reprinted figures

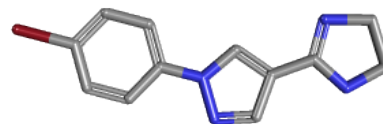
Figures S1 and S2 reprint the main-text protocol plate and the $n=500$ wall so the supplement can be read alone. Listing 1 is reprinted with the same live listing+framed layout (PubChem 3D, CID 57398578).



Figure S1: Reprint of main-text Figure 1 (fig1_leadoverview). Generate 249/500 (49.8%) recall in top 3; select 204/249 (81.9%) verified | pool ($49.8\% \times 81.9\% = 40.8\%$ verified). Wall: 204 verified / 45 misranked / 251 never proposed.

Listing 1. Example IRExp JSON record (commercial DoR; reprint of main-text Listing 1; ILJNJNKEOAREX) — PubChem 3D CID 57398578. SMILES and InChIKey are resource fields, not solver payload.

```
{
  "id": "ILJNJNKEOAREX-UHFFFAOYSA-N",
  "formula": "C12H11BrN4",
  "ir_bands_cm-1": [3144.0, 3110.0, 3066.0, 2931.0,
    2863.0, 1632.0, 1589.0, 1565.0,
    1485.0],
  "ir_source": "experimental",
  "source_doi": "PMC:12566713",
  "pmcid": "PMC12566713",
  "h_nmr": "8.83 (s, 1H), 8.04 (s, 1H), 7.81 (d, J = 8.8
    Hz, 2H),
    7.72 (d, J = 8.8 Hz, 2H), 3.56 (s, 4H)",
  "c_nmr": "157.5, 140.3, 138.4, 132.4, 127.5, 120.4,
    119.0, 116.4, 49.1",
  "smiles": "BrC1CCC(-N2CC(C3=NCCN3)CN2)CC1",
  "inchikey": "ILJNJNKEOAREX-UHFFFAOYSA-N",
  "has_structure": true,
  "license": "CC-BY",
  "license_pool": "commercial"
}
```



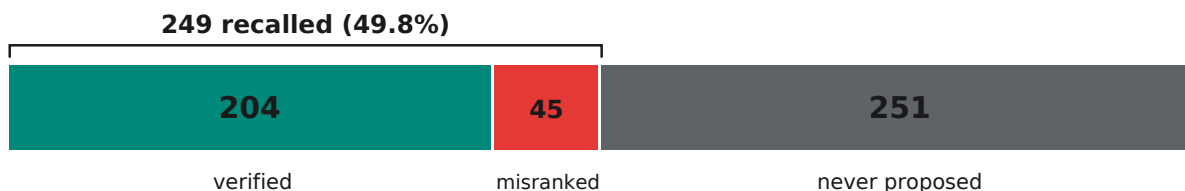


Figure S2: Reprint of the $n=500$ forward-verify wall (`fig.wall.diagnostic`): 204 verified / 45 mis-ranked / 251 never proposed. Source: `spectro-agent data/fverify_n500/WALL_n500.md`. Generation remains 227/500 top-1 and 249/500 recall.

3 Blind protocol

3.1 Solver-facing contract

Each IR spectra-Bench item is a four-field JSON object. The solver sees *only*

```
{qid, formula, ir_bands_cm-1, h_nmr, c_nmr}.
```

No name, SMILES, InChIKey, scaffold, atom-type hint, retrieved spectrum, licence stamp, or PMC id enters the prompt (main-text Listing 1, adapted from one IReXP record). Gold constitution is scored offline by the InChIKey-14 indicator (§4).

Difficulty is declared *a priori* by RDKit ring analysis: *simple* iff ≤ 2 rings, no fused/spiro/bridgehead system, ≤ 22 heavy atoms; else *complex*. The $n=500$ headline is near-50/50 (248/252); the eligible IReXP corpus is 17.5% simple / 82.5% complex. Corpus-reweighted extras on a validate-clean subset are in §5 and are not $n=500$ figures.

3.2 Pre-registration

The following files live in the companion spectro-agent tree. They record the frozen roster and the expand-500 campaign.

- $n=500$ roster: frozen `qid` lists in the code release. Sampler seeds for early draws were not captured; the released roster is fixed and reusable but not exactly re-drawable.
- expand-500 campaign: `docs/EXPANSION_PREREGISTRATION_500.md` (seed 2026500; 115/115 simple/complex). 0 collisions with any prior round. Six ^{13}C -overread flags known *before* any solve: R26, R31, R102, R105, R107, R138.

3.3 Stopping and pooling rules

- The $n=500$ roster is the pre-specified frozen list (`docs/headline500_expand200_qids.json` plus the committed deposits), not an outcome-selected subset.
- `score_pooled.py --expand-500` refuses a missing `predictions2.jsonl` or a withheld key.
- `answers2.jsonl` is absent from both working trees.
- Keys were restored only under `/tmp/blind/_key/` for the 2026-09-20 score pass and then deleted.
- `validate_benchmark.py` was not re-run on that pass (the existing 224-qid snapshot is used).
- `collect_round.py` was not run; 230 `raw/single_R*.json` files were left untouched.

- `forward_verify_main.py` score was run on expand-500 (§8): 103/230 verify vs 129/230 self. Official prep was not re-run.
- Bootstrap intervals were not computed for expand-500.

3.4 Allowed tools

RDKit formula / parse check only. No web search. No retrieved spectrum. No thinking tier on the core protocol. A thinking-tier consumer arm is labelled as such in the score notes and is not interchangeable with the no-thinking protocol.

A deposit is JSONL of $\{\text{qid}, \text{candidates: } [\text{SMILES}, \dots]\}$ under a stated budget. Headline budget: ≤ 3 ranked SMILES per qid. Self-ranking and forward-verified re-ranking are reported separately.

4 Scoring identities

These identities match main-text Eqs. (1)–(7).

A candidate $\hat{y}$ matches gold y^* at constitution iff

$$s(\hat{y}, y^*) = \mathbf{1}[\text{IK}_{14}(\hat{y}) = \text{IK}_{14}(y^*)]. \quad (\text{S1})$$

IK_{14} is the first 14 characters of the RDKit InChIKey (connectivity layer). Full stereo-sensitive InChIKey is reported only on validate-clean $n=295$ (§5).

With ranked pool $\mathcal{C}_i = (c_{i1}, \dots, c_{ik})$, $k \leq 3$,

$$\text{top-1} = \frac{1}{n} \sum_i s(c_{i1}, y_i^*), \quad (\text{S2})$$

$$\text{recall}@k = \frac{1}{n} \sum_i \mathbf{1}[\exists j \leq k : s(c_{ij}, y_i^*) = 1], \quad (\text{S3})$$

$$\text{prec.}|\text{recall} = \frac{\sum_i s(c_{i1}, y_i^*)}{\sum_i \mathbf{1}[y_i^* \in \mathcal{C}_i]}. \quad (\text{S4})$$

These are the three reportable numbers. They factorise as

$$\text{top-1} = (\text{recall}@k) \times (\text{prec.}|\text{recall}) \quad (\text{S5})$$

when the same pool is used for both stages. The rates have different denominators and must not be differenced. Self-ranking $\text{prec.}|\text{recall}$ on $n=500$ is 227/249 (91.2%), which is not forward-verify precision.

Forward-verify re-ranks by symmetric chamfer on ^{13}C peak sets $P, Q \subset \mathbb{R}$

$$d(P, Q) = \frac{1}{|P|} \sum_{p \in P} \min_{q \in Q} |p - q| + \frac{1}{|Q|} \sum_{q \in Q} \min_{p \in P} |q - p|, \quad (\text{S6})$$

where $\hat{P}(c)$ is predicted *blind* to the observed list P_{obs} . An instrumented diagnostic on a protocol slice is 58/65 (§8). The $n=500$ forward-verify wall is 204/45/251 (§8; `data/fverify_n500/WALL_n500.md`). The expand-500 official arm score is 103/230 verify vs 129/230 self (§8).

A pool estimator on an index set $\mathcal{S}$ is

$$\hat{p}_{\mathcal{S}} = \frac{1}{|\mathcal{S}|} \sum_{i \in \mathcal{S}} s(c_{i1}, y_i^*). \quad (\text{S7})$$

The ICLR headline uses $\mathcal{S}$ of size 500 (frozen qid list). The $n=519$ `score_pooled.py` print (243/519) is not used.

Confidence intervals on earlier validate-clean subsets are bootstrap 95% (`score_main.boot`, seed 0, 2000 resamples). Paired model differences use McNemar with Holm correction. Confidence intervals are not reported for $n=500$ or expand-500.

5 Validate-clean extras

Source: `docs/POOLED_HEADLINE_2026-09-16.md`. These rows are not the $n=500$ headline. Self-ranking precision | recall on $n=500$ is 227/249 (91.2%).

¹³C-overread flags (R12, R22, R25, R82, R91) were marked before subset totals (`scripts/validate_benchmark.data/benchmark_expand/clean_qids.json`).

Table S2: Validate-clean extras. Same scorer; not the $n=500$ headline.

metric	overall	simple	complex
scaffold-level (best Tanimoto ≥ 0.45)	64%	80%	48%
mean best Tanimoto	0.66	0.79	0.53
full InChIKey (stereo) top-1	93/295 (31.5%) [26–37]	—	—
full InChIKey recall	108/295 (36.6%) [31–42]	—	—
corpus-reweighted top-1	26.5% [21–32]	—	—
corpus-reweighted recall	31.3% [25–37]	—	—

Table S3: Size bins on validate-clean $n=295$.

heavy atoms	n	top-1	recall
≤ 15	53	36/53 (67.9%)	39/53 (73.6%)
16–25	152	65/152 (42.8%)	75/152 (49.3%)
> 25	90	15/90 (16.7%)	16/90 (17.8%)

On an instrumented slice, of 137 analysable top-1 misses, 76.6% are constitutional isomers of the true structure; 22.6% share the true Murcko scaffold. Named-ring annotation on that slice is rare and does not drive the result.

6 expand-500 generation

Source: `spectro-agent data/benchmark_expand_500/NIGHT_POOL_2026-09-20.md` (copied to `docs/NIGHT_POOL_2026-09-20.md`). The key was restored only for `score2` and then deleted. `answers2.jsonl` is not in the tree.

Mean best Tanimoto (`score2` footer, all 230): simple 0.863; complex 0.645. Flagged qids (in 230, out of 224): R26, R31, R102, R105, R107, R138. Deposits: 230/230; `predictions2.jsonl` 230 lines, 690 candidates. Unique (formula, IR, ¹³C) vs `irexp_resolved.jsonl.gz`: 230/230; 0 prior-round InChIKey-14 collisions. This arm used a thinking-tier consumer protocol and is not interchangeable with the no-thinking core protocol.

Verbatim `score2` footer:

overall recovered (top-3): 138/230 (60%) top-1 exact: 129/230 (56%)

Table S4: expand-500 generation (score2, InChIKey-14). No confidence intervals.

set	<i>n</i>	top-1	recall@3
all deposited Opus	230	129/230 (56.1%)	138/230 (60.0%)
simple	115	82/115 (71%)	87/115 (76%)
complex	115	47/115 (41%)	51/115 (44%)
validate-clean	224	127/224 (56.7%)	135/224 (60.3%)
simple	110	81/110 (74%)	85/110 (77%)
complex	114	46/114 (40%)	50/114 (44%)

simple : recovered 87/115 (76%) top1 82/115 (71%) meanBestTani 0.863
 complex : recovered 51/115 (44%) top1 47/115 (41%) meanBestTani 0.645

7 Headline generation

The estimator is Eq. S7. The paper headline is $n=500$. Source: docs/HEADLINE.N500.2026-09-20.md (spectro-agent 48fcd30). Roster: frozen qid lists in the code release (docs/headline500_expand200_qids.json)

Table S5: Headline generation on the frozen $n=500$ roster. Confidence intervals are not reported. This table is generation only.

set	<i>n</i>	top-1	recall@3
paper headline	500	227/500 (45.4%)	249/500 (49.8%)

Strata on the $n=500$ headline (same InChIKey-14 rule): simple 161/248 (65%) top-1 / 172/248 (69%) recall; complex 66/252 (26%) / 77/252 (31%). Self-rank precision | recall: 227/249 (91.2%). The $n=519$ pooled print (243/519) is not the headline. The $n=500$ forward-verify wall is 204/45/251 (§8).

8 Forward verification

8.1 $n=500$ unified wall

Source: spectro-agent data/fverify_n500/WALL_n500.md (scripts/fverify_n500_wall.py). Classification matches the sidecar: **verified** = recalled and chamfer-ranked first; **misranked** = recalled, distractor first; **never-proposed** = true structure not in the candidate set. The paper wall is the $n=500$ row. Arm rows below are not the headline.

Table S6: $n=500$ forward-verify wall, copied from data/fverify_n500/WALL_n500.md. No confidence intervals. Arm rows are not the headline.

arm	<i>n</i>	verified	misranked	never-proposed
locked protocol slice	194	58	7	129
Opus expansion	106	56	12	38
expand-500 cut	200	90	26	84
$n=500$ wall	500	204	45	251

Recalled = 204 + 45 = 249/500, matching committed generation recall. Main-text Figure fig-wall

reprints this 204/45/251 wall. Figure 1 reports the same $n=500$ decomposition (249/500 recall $\times$ 204/249 verified | pool = 40.8% verified).

8.2 Instrumented protocol slice

Source: `data/diagnosis.json / scripts/forward_verify_all.py`. This is a protocol-slice diagnostic, not a second $n=500$ wall.

Table S7: Instrumented forward-verify diagnostic on a protocol slice. Not the $n=500$ wall (204/45/251).

	60-compound arm	protocol slice
generation recall	19/60 (32%)	65 (34%)
top-1, solver self-ranking	14/60 (23%)	55 (28%)
top-1, forward-verified re-ranking	16/60 (27%)	58 (30%)
precision recall — fverify	16/19 (84%)	58/65 (89%)
multi-candidate only — fverify	10/13 (77%)	30/37 (81%)
derangement floor on $n=37$	—	54.0%; fverify +27.1 points
McNemar vs self-rank	—	$p=0.55$ (7 gained, 4 lost)

Self-ranking 227/249 on $n=500$ is generation, not a substitute for 204/45/251.

8.3 expand-500 official score

Source: `spectro-agent data/fverify_expand_500/STATUS.md, results.txt, and NIGHT_POOL_2026-09-20 Blind 13C campaign over 681 unique candidate SMILES (f1-f41 all present)`. Official `forward_verify_main.py score()` ran 2026-09-20 (`is_true` filled under `/tmp` only; committed `candidates.jsonl` still omits it; official prep was not re-run). Verify does not beat self-rank. Headline generation remains $n=500$ (227/500, 249/500). This arm score does not replace `data/fverify_n500/WALL_n500.md`.

Table S8: expand-500 official chamfer score. Arm score, not the $n=500$ wall and not the headline.

set	n	recall	self top-1	verify top-1	verify recall
all deposited	230	138/230 (60.0%)	129/230 (56.1%)	103/230 (44.8%)	103/138 (74.6%)
validate-clean	224	135/224 (60.3%)	127/224 (56.7%)	103/224 (46.0%)	103/135 (76.3%)

Coverage: 41/41 batches; 681/681 unique SMILES; 230/230 qids fully covered. Multi-candidate only: all-230 self 128/137, verify 102/137; clean-224 self 126/134, verify 102/134. `forward_verify_all.py` was not run (no `expand-500 diagnosis.json`).

HOSE-style lookup on the protocol-slice candidate sets ties self-ranking; a modest GNN reaches 59/65 (91%) vs the LLM verifier’s 58/65 (89%). Deranging observed ¹³C pairings collapses precision ($p=0.001$).

9 Worked cases

The three locked-slice modes in the main appendix, plus additional panels that use already-reported numbers. Expansion qids reuse R22/R25. No SMILES are invented for R06/R26; only v3-R25 molecules are drawn (Figure S3).

Table S9: Three-mode case grid. Same integers as main-text Table `tab:cases`.

mode	id / formula	notes
Proposal miss	main-R06 / $C_{29}H_{30}FNO_2$	True acridinedione never in the 3-candidate pool (Tanimoto-0.90 regioisomer: 4-fluorophenyl / phenyl swapped). IR 1641, 1510 cm^{-1} ; ^{13}C 195.7. Verifier unused.
Verify-save	v3-R25 / $C_{10}H_{14}N_2O$	True 2-pyridyl picolinamide at self-rank 3; self-rank 1 is 3-pyridyl. Forward-verify recovers (chamfer 0.42 vs 1.30 ppm). IR 1679 cm^{-1} . Figure S3.
Verify-fail	v3-R26 / $C_{14}H_{11}N_3O_3$	True 3-nitrophenyl dihydroquinazolinone in pool (self-rank 2); 2-nitro ranked first. Chamfer 1.35 vs 1.36 ppm. IR 1652, 1530, 1352 cm^{-1} .

9.1 Proposal miss — main-R06 / $C_{29}H_{30}FNO_2$

True acridinedione never enters the 3-candidate pool (Tanimoto-0.90 regioisomer: 4-fluorophenyl / phenyl swapped). IR 1641, 1510 cm^{-1} ; ^{13}C 195.7. Verifier unused. Re-ranking cannot recover a structure that was never proposed (Eq. S5).

9.2 Verify-save — v3-R25 / $C_{10}H_{14}N_2O$

True 2-pyridyl picolinamide at self-rank 3; self-rank 1 is 3-pyridyl. Forward-verify recovers (chamfer 0.42 vs 1.30 ppm, Eq. S6). IR 1679 cm^{-1} . Main-text Figure 5; SI Figure S3.

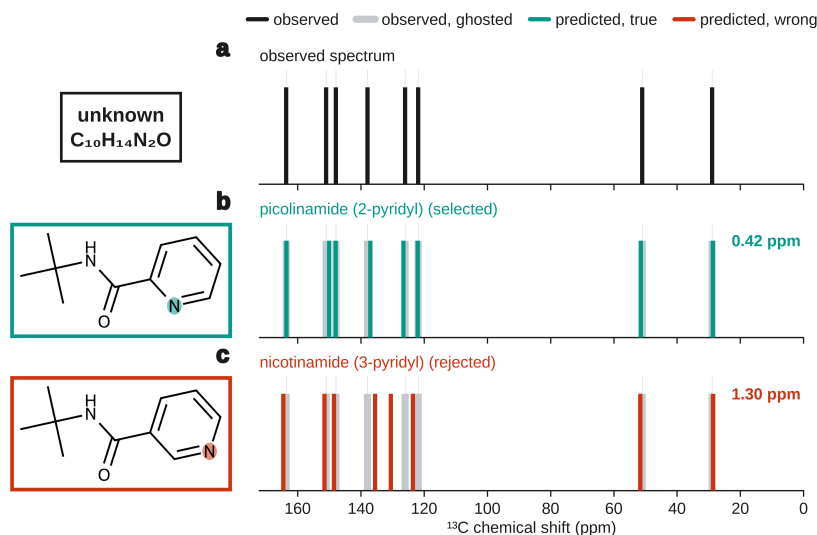


Figure S3: Locked v3-R25 verify-save (main-text Figure 5). Self-rank prefers 3-pyridyl; forward ^{13}C recovers 2-pyridyl (chamfer 0.42 vs 1.30 ppm).

9.3 Verify-fail — v3-R26 / $C_{14}H_{11}N_3O_3$

True 3-nitrophenyl dihydroquinazolinone in pool (self-rank 2); 2-nitro ranked first. Chamfer 1.35 vs 1.36 ppm. IR 1652, 1530, 1352 cm^{-1} . Ranking can fail when predicted ^{13}C lists are nearly tied (McNemar $p=0.55$; 7

gained, 4 lost on the protocol slice).

9.4 Listing 1

Main-text Listing 1 is the live IReXp JSON record ($\text{C}_{12}\text{H}_{11}\text{BrN}_4$; InChIKey `ILJNJUNKEOAREX-UHFFFAOYSA-N`; PubChem CID 57398578). SMILES, InChIKey, and licence fields are resource metadata. Bench solvers see only formula and printed peak lists. Gold constitution is scored only by Eq. S1. The listing is reprinted above with the same `lstlisting+framed` layout. Locked v3-R25 remains the mechanism worked case (main-text Figure 5).

9.5 Additional reported panels

Formula-only control ($n=60$). Formula only: 3/60 (5%) top-1 and 3/60 (5%) recall. Formula + IR + ^1H + ^{13}C : 14/60 (23%) / 19/60 (32%). Outcomes perfectly nested (McNemar $p=0.001$).

Generate-wide (60-compound arm). Ten solver agents, up to six candidates each, pooled then re-ranked: recall 32% $\rightarrow$ 42%, top-1 23% $\rightarrow$ 30% (McNemar $p=0.34$); verification precision falls 84% $\rightarrow$ 72%.

IReXp-fine-tuned generator (locked slice). Pooled with Claude: recall 33.5% $\rightarrow$ 54.1%, top-1 28.4% $\rightarrow$ 35.1% (McNemar $p=0.015$). Training is the intervention that has been observed to move $P(\text{true} \in \text{pool})$.

Battery-electrolyte subset. $n=46$ literature electrolyte chemistry (not operando): 26% top-1, recall-bound.

Four-model Claude ladder ($n=24$). Haiku $\subset$ Sonnet $\subset$ Opus $\subset$ Fable is capability-sensitive but underpowered for adjacent ranks, with a protocol asymmetry.

Difficulty variation. Pre-registered draws differ in difficulty. The headline remains $n=500$ (Eq. S7).

10 Prompt skeletons and deposit schema

10.1 Headline solver-facing skeleton

Wrapping instruction text was not captured on the headline Claude arm (main-text Limitations (i)). Only the per-compound four-field payload is released. The skeleton below is the contract, not a recovered verbatim system prompt.

```
You are given only:
  formula          (as from HRMS)
  ir_bands_cm-1    (author-transcribed experimental list)
  h_nmr, c_nmr     (printed shift lists)
Return <= 3 ranked SMILES.
No name, no SMILES hint, no scaffold, no retrieved spectrum.
Check molecular formula with RDKit before emitting a candidate.
```

Listing 1 payload (solver-facing fields only; gold held out):

```
{
  "formula": "C12H11BrN4",
  "ir_bands_cm-1": [3144.0, 3110.0, 3066.0, 2931.0,
                    2863.0, 1632.0, 1589.0, 1565.0, 1485.0],
  "h_nmr": "8.83 (s, 1H), 8.04 (s, 1H), 7.81 (d, J = 8.8 Hz, 2H), ...",
```

```

    "c_nmr": "157.5, 140.3, 138.4, 132.4, 127.5, 120.4, 119.0, 116.4, 49.1"
}

```

10.2 Forward-verify skeleton

For each candidate SMILES, predict a ^{13}C peak list *without* seeing the observed list. Re-rank by $d(\hat{P}(c), P_{\text{obs}})$ (Eq. S6). The instruction dispatched to the forward-prediction sub-agents was not captured; only the candidate SMILES and the resulting numeric lists are on disk for the protocol slice.

10.3 Cross-vendor harness

Verbatim prompts in the spectro-agent tree (ESI S2) are the cross-vendor harness prompts, not the missing headline Claude wrapper. External submissions must document model version, prompt, and tool access (docs/LEADERBOARD.md).

10.4 Deposit schema

```

{"qid": "R01", "candidates": ["SMILES_rank1", "SMILES_rank2", "SMILES_rank3"]}

```

Score locally (spectro-agent):

```

python scripts/score_submission.py --predictions my_run.jsonl --name "YourModel-1.0"
python scripts/score_pooled.py --stereo      # validate-clean stereo extras
python scripts/forward_verify_all.py         # protocol-slice diagnostic

```

11 Model cards

11.1 Headline model card

Table S10: Headline solver. Unrecorded fields are left blank rather than guessed.

item	value
headline solver	Claude Opus, consumer subscription
thinking tier (core protocol)	none; a labelled thinking-tier arm exists
snapshot / seed / temperature / top_p / top_k / max_tokens	not exposed; not recorded
candidate budget	≤ 3 SMILES / qid
tool access	RDKit formula / parse check only; no web search
fine-tuning	none on the headline arm
paid API	no, for the core protocol
hardware	consumer harness; no cluster job for inference
scoring hardware	commodity CPU; RDKit InChIKey-14
cross-vendor arm	$n=60$; Grok 4.6, Gemini 3.7 Flash, GPT-5.6 Sol
Claude mean candidates on that arm	2.20 (others 3.00) — recall rankings approximate
expand-500 solver	Claude Opus thinking-tier (declared; not headline)
Fable incomplete arm	deposits not scored; not pooled
scoring scripts	scripts/score_pooled.py, score_main.py, benchmark_v2.py score2

Table S11: Cross-vendor decomposition. Recall and precision use different denominators.

model	generation recall	prec. recall	multi-candidate only
Claude Opus	19/60 = 32%	16/19 = 84%	10/13 = 77%
Grok 4.6	32/60 = 53%	20/32 = 62%	20/32 = 62%
Gemini 3.7 Flash	30/60 = 50%	22/30 = 73%	22/30 = 73%
GPT-5.6 Sol	25/60 = 42%	17/25 = 68%	16/24 = 67%

11.2 Cross-vendor card ($n=60$)

Verification precision exceeds generation recall in every arm. Bootstrap separates the paired gap for Claude (+52.5 points), GPT-5.6 Sol (+26.3) and Gemini (+23.3); Grok’s gap (+9.2) is directional. A clean-clone control for Grok bounds key leakage (McNemar $p=0.39$). Reasoning-effort tier per vendor arm was not recorded.

11.3 Literature decomposition

Any system reporting top-1 = a and top- k = b implies recall $\geq b$ and conditional precision $\leq a/b$. Script: `scripts/literature_decomposition.py`.

Table S12: Selected literature decomposition rows (as in main-text Table 6).

system (n)	top-1	top- k (k)	recall	prec.
this work, solver alone (500)	45.4%	49.8% (3)	49.8%	91.2%
this work + forward-verify (slice)	29.9%	33.5% (3)	33.5%	89.2%
NMR-Solver, literature (450)	52.9%	67.3% (10)	$\geq 67.3\%$	$\leq 78.6\%$
NMRAgent, literature (450)	61.6%	70.0% (10)	$\geq 70.0\%$	$\leq 88.0\%$
Espejo Morales, education (236)	80.9%	90.0% (5)	$\geq 90.0\%$	$\leq 89.9\%$
Espejo Morales, industrial (34)	20.6%	29.1% (5)	$\geq 29.1\%$	$\leq 70.9\%$
Alberts, 6–13 atoms, NIST IR (3,455)	63.8%	84.0% (10)	84.0%	75.9%
SpecX, scaffold split, simulated (99,439)	29.7%	50.6% (10)	50.6%	58.7%

Three groups changed only the data, and recall carried 68% (NMR-Solver simulated $\rightarrow$ real), 70% (SpecX random $\rightarrow$ scaffold) and 83% (Espejo education $\rightarrow$ industrial) of each collapse. The solver-alone precision column on $n=500$ is self-ranking (227/249); the forward-verify row is the instrumented protocol slice (58/65).

12 Companion resource and anonymity

IRSpectra-Bench is the blind peak-list benchmark and the propose $\gg$ verify diagnosis. IRExp is a companion *Scientific Data* Data Descriptor (in preparation): 121,233 records; 43,060 structure-linked; 33,201 full IR+ ^1H + ^{13}C +structure quadruples. This supplement cites that resource and does not re-present Methods, Data Records, or licence-pool narrative.

Double-blind review copy: <https://anonymous.4open.science/r/peaklist-corpus-review-10C4/> Named archival hosting is restored at camera-ready. Fine-tunes must use `data/train_no_bench.jsonl.gz` so benchmark InChIKeys are withheld.

Listing 1 is the IRExp-style JSON+molecule plate in the ICLR PDF.

13 Reproducibility

Frozen questions and predictions live in the spectro-agent release; this repository holds the ICLR manuscript.

The $n=500$ generation totals 227/500 and 249/500 regenerate from `docs/HEADLINE_N500_2026-09-20.md`.

The $n=500$ wall 204 verified / 45 misranked / 251 never-proposed is recorded in `spectro-agent data/fverify_n500/WAL`.

The protocol-slice diagnostic is regenerated by `python scripts/forward_verify_all.py`.

To rescore expand-500, restore `/tmp/blind/_key/benchmark_expand_500.answers2.jsonl.withheld`, run `benchmark_v2.py score2 --outdir data/benchmark_expand_500`, and re-withhold the `key.answers2.jsonl` must not be committed. The official expand-500 chamfer score is 103/230 verify vs 129/230 self; official prep was not re-run.

Companion bibliographic entries are anonymous in the review copy; the dataset URL in the PDF is the anonymous.4open review copy. Fine-tunes should use `data/train_no_bench.jsonl.gz`.

Inference is not bit-exact: no pinned snapshot, temperature, or seed was recorded. Distributional agreement of frozen deposits is the intended claim. Uncaptured artefacts: headline Claude wrapper text; dated model snapshots; decoding parameters; main-round sampler seeds; closed-book audit transcripts (`docs/SUBMISSION.md`).

Deferred analyses: expert-chemist audit; leave-one-modality-out; parser extraction-recall audit (companion Data Descriptor); on-bench scores for SpectraLLM, IR-Agent, Spectro, NMIRacle, Alberts, and CASE.

Manuscript compile: `python3 scripts/build_pdf.py (main); python3 scripts/build_si.py` (this supplement). Engine: tectonic 0.17.0 or pdfLaTeX / XeLaTeX + BibTeX.